

Star Sensor Star Pattern Recognition in Lost-in-Space (LIS) Mode for Space Applications

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Abstract—Star trackers are critical for spacecraft attitude determination due to their high accuracy in estimating orientation. This capability is paramount during Lost-in-Space (LIS) operations, where no prior attitude information is available. However, existing LIS star identification methods implemented on resource-constrained embedded platforms are often degraded by centroiding errors, false detections, and ambiguous geometric matches. This study proposes a multilayer geometric voting framework for autonomous LIS star identification and attitude determination. The methodology utilizes the Hipparcos Star Catalog to construct a star-pair database and processes real astronomical images through threshold-based detection, primary geometric voting, and a novel secondary consistency verification stage. The system was implemented on a Commercial Off-The-Shelf (COTS) Raspberry Pi 5 platform and evaluated using calibrated astronomical images with World Coordinate System (WCS) validation. Results demonstrated that the proposed framework successfully identified stars across multiple test images, yielding high-confidence matches consistent with WCS-derived celestial coordinates. The QUEST and Davenport q-method algorithms generated valid quaternion-based attitude solutions with negligible discrepancy. Ultimately, the embedded implementation establishes a highly reliable, low-cost foundation suitable for future CubeSat and small satellite applications.

Index Terms— Attitude determination, CubeSat, geometric voting, Lost-in-Space (LIS), Raspberry Pi, star tracker.

I. INTRODUCTION

THE evolution of spacecraft navigation has rapidly shifted from ground-based tracking and manual celestial navigation toward fully autonomous onboard systems. At the heart of this autonomy is the star tracker, an optoelectronic instrument that determines spacecraft attitude with arc-second level accuracy by identifying stars within its Field of View (FOV). Historically, conventional space-grade star trackers relied on highly specialized, radiation-hardened hardware. While reliable, these systems are associated with substantial development and procurement costs, limiting their accessibility for smaller missions.

In recent years, the emergence of the New Space paradigm has accelerated the adoption of Commercial Off-The-Shelf (COTS) technologies. Advances in embedded computing platforms, such as the Raspberry Pi 5, have enabled the development of lower-cost and more compact star tracker systems suitable for resource-constrained spacecraft like CubeSats and

nanosatellites. However, achieving reliable Lost-in-Space (LIS) performance on such hardware remains a significant challenge. Current LIS star identification systems often require computationally expensive blind catalog searching and geometric pattern matching against large onboard star databases, leading to high processing overhead.

Furthermore, existing star identification algorithms are highly sensitive to centroiding noise, false star detections, and sensor uncertainties, which can degrade identification accuracy under practical imaging conditions. While recent studies have proposed independent improvements in centroiding or voting-based identification techniques, a critical research gap emerges: very few methods are evaluated under combined, interacting disturbances (such as overlapping sensor noise, sparse star fields, and motion blur). This study addresses these limitations by developing a complete, low-cost embedded LIS pipeline featuring a robust multilayer geometric voting framework with a secondary consistency verification stage to systematically eliminate false positives before attitude estimation.

II. LITERATURE REVIEW

Early LIS algorithms relied on classical geometric invariants, such as triangle and multi-star configurations. While computationally efficient, these methods deteriorate sharply under centroiding noise, magnitude variations, or missing detections. To optimize processing, geometric hashing and multi-dimensional indexing techniques like the K-vector approach were introduced. The K-vector enables rapid candidate retrieval based on angular separation but can return ambiguous candidate pairs, necessitating secondary validation to reject inconsistent hypotheses.

More recently, voting-based algorithms have improved robustness by accumulating statistical evidence from multiple star relationships. A prominent COTS-based hardware benchmark is StarberrySense, a low-cost Raspberry Pi star sensor utilizing localized geometrical voting [1]. While successfully flight-tested, orbital data exposed vulnerabilities: spacecraft tumbling induced star trailing and centroid errors, which compromised the single-stage geometrical voting process. These operational vulnerabilities directly justify the core motivation of this research: accelerated database indexing must be coupled with an independent, multi-layered logical verification stage to guarantee identification reliability under real-world imaging conditions before computing the attitude

matrix.

III. METHODOLOGY

The algorithmic architecture of the proposed autonomous star tracker system is structured into four sequential operational phases designed to execute on an embedded processor: reference catalog preparation, real-time image processing and centroid extraction, multi-layer geometric pattern voting, and deterministic attitude determination solvers.

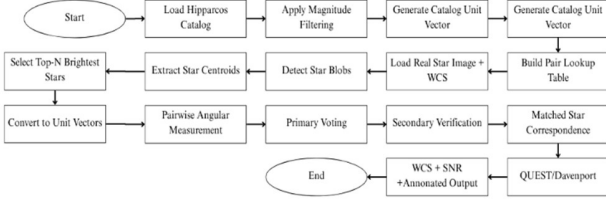


Fig. 1. Block diagram illustrating the catalog caching, real-time image processing, and multilayer voting architecture.

A. Catalog Preparation

The foundational reference coordinate framework is constructed offline from the Hipparcos Star Catalog. To optimize the memory constraints of a low-cost, Commercial Off-The-Shelf (COTS) embedded computer, the catalog is filtered dynamically by establishing a limiting apparent visual magnitude threshold ($m_{\text{lim}} \leq 6.5$). This ensures that only visible, high-intensity guide stars are preserved. The celestial spherical coordinates of right ascension (α_i) and declination (δ_i) for each star are mapped into 3D inertial unit vectors $\mathbf{r}_i$ within the Earth-Centered Inertial (ECI) reference frame using the trigonometric relation:

$$\mathbf{r}_i = \begin{bmatrix} \cos \delta_i \cos \alpha_i \\ \cos \delta_i \sin \alpha_i \\ \sin \delta_i \end{bmatrix} \quad (1)$$

To minimize runtime execution overhead during Lost-in-Space (LIS) blind searches, an angular distance lookup table is constructed. The angular separation (θ_{ij}) between every possible star-pair within the sensor's localized Field of View (FOV) is pre-computed via the vector dot product: $\theta_{ij} = \cos^{-1}(\mathbf{r}_i \cdot \mathbf{r}_j)$. These separations are indexed using a highly optimized search array, enabling instantaneous, sub-millisecond database queries.

B. Image Processing and Centroiding

When the system captures a real-world night-sky astronomical image, it undergoes a two-stage segmentation pipeline to isolate authentic stellar sources from background thermal noise and sensor artifacts. First, a statistical global detection threshold (T) is dynamically calculated to accommodate variations in background illumination:

$$T = \mu + 5\sigma \quad (2)$$

C. Image Processing and Centroiding

Before running the pattern recognition algorithm, the raw pixel centroids (x, y) must be converted into physical direction vectors in the 3D camera body frame. The pixel measurements are first shifted relative to the optical principal center (c_x, c_y) to determine the localized offsets: $u = x - c_x$ and $v = y - c_y$. To mitigate structural spatial inaccuracies caused by lens aberrations, second-order geometric distortion corrections are applied along both axes using Simple Imaging Polynomial (SIP) coefficients derived from astrometric calibration data:

$$\Delta u = A_{02}v^2 + A_{11}uv + A_{20}u^2 \quad (3)$$

$$\Delta v = B_{02}v^2 + B_{11}uv + B_{20}u^2 \quad (4)$$

The mathematically corrected pixel positions are resolved as $u_c = u + \Delta u$ and $v_c = v + \Delta v$. Utilizing the calibrated pixel focal length (f_{px}), these positions are projected into a normalized 3D camera-frame body vector $\mathbf{b}_i$:

$$\mathbf{b}_i = \frac{1}{\sqrt{x_i^2 + y_i^2 + 1}} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} \quad (5)$$

where $x_i = u_c/f_{px}$ and $y_i = -v_c/f_{px}$, with the negative inversion matching the physical orientation of the image sensor array.

D. Multilayer Geometric Voting

The core contribution of this system is a two-stage hierarchical voting process designed to confidently establish identities for the observed stars while completely discarding ambiguous structural configurations.

1) Primary Star-Pair Voting

The algorithm evaluates the observed angular separations (θ_{mn}^{obs}) between all combinations of the extracted body vectors: $\theta_{mn}^{\text{obs}} = \cos^{-1}(\mathbf{b}_m \cdot \mathbf{b}_n)$. The system queries the pre-computed database for catalog pairs (k, l) whose pre-indexed distances fall within a strict operational error tolerance window ($\epsilon = 20$ arcseconds):

$$|\theta_{mn}^{\text{obs}} - \theta_{kl}^{\text{cat}}| \leq \epsilon \quad (6)$$

When a match is found, primary votes V_1 are incremented for both observed indices against their corresponding catalog candidates. The highest-voted matches form an initial candidate identity list.

2) Secondary Verification / Second-Level Voting

To handle crowded star fields or high centroiding noise where primary voting returns false matching trends, a

secondary logical verification layer is executed. The system extracts the top candidate identities and evaluates their mutual multi-directional geometry. A secondary vote V_2 is only awarded if the physical distance between two assigned catalog stars matches the observed distance between their respective camera centroids within a localized tolerance ($\tau_2 = 0.01$ radians). A star must achieve a score of $V_2 \geq 2$ to be accepted into the final verification matrix. If a sparse field or high disturbance causes the candidate stars to fail this mutual distance test, the system registers a "Safe Rejection" to halt navigation before corrupted vectors are passed forward.

E. Attitude Determination

Verified correspondences $(\mathbf{b}_i, \mathbf{r}_i)$ are passed to the Quaternion Estimator (QUEST) and Davenport's q-method solvers. Both solve Wahba's problem by constructing the attitude profile matrix B :

$$B = \sum w_i \mathbf{b}_i \mathbf{r}_i^T \quad (7)$$

1) QUEST Algorithm

The Quaternion Estimator (QUEST) reformulates the profile matrix into a localized characteristic polynomial to solve for the maximum eigenvalue ($\lambda_{\max}$) without relying on computationally intensive iterations. The optimal spacecraft attitude quaternion $\mathbf{q}_{\text{opt}}$ is computed directly through a series of linear vector multiplications, ensuring extremely fast processing speeds on embedded devices.

2) Davenport's q-Method

To validate the reliability of the QUEST output, Davenport's q-method explicitly constructs a symmetric 4×4 design matrix K using the elements of the profile matrix B :

$$K = \begin{bmatrix} \text{Tr}(B) & \mathbf{Z}^T \\ \mathbf{Z} & S - \text{Tr}(B)\mathbf{I} \end{bmatrix} \quad (8)$$

where $\mathbf{Z}$ represents the anti-symmetric components of B , $S = B + B^T$, and $\mathbf{I}$ is the identity matrix. The optimal navigation quaternion is extracted via direct eigenvalue decomposition:

$$\mathbf{q}_{\text{opt}} = \text{eigvec}_{\max}(K) \quad (9)$$

The resulting optimal quaternion vector is finally converted into boresight right ascension (α) and declination (δ) pointing coordinates to facilitate navigation assessment.

IV. RESULTS AND DISCUSSION

The proposed multi-layer geometric voting framework and attitude determination pipeline were fully implemented and evaluated on a Commercial Off-The-Shelf (COTS) Raspberry Pi 5 platform. To rigorously test the robustness of the system under real-world conditions, four distinct astronomical test images were selected from a calibrated database. These test cases represent diverse celestial environments, varying fields of view (FOV), differing star densities, and various signal-to-noise ratio (SNR) challenges. The specific parameters and conditions of each test image are structured and summarized in Table I.

TABLE I
SUMMARY OF EXPERIMENTAL TEST IMAGES CONDITIONS

Test Image	Image Condition	FOV (deg)	Extracted Centroids	Target Stars
Image 1	Clean / Moderate FOV	9.33	33	20
Image 2	Wide FOV Stress Case	29.19	3283 (with noise)	20
Image 3	Lower SNR / Crowded	Standard	Dense Cluster	20
Image 4	Limited-Star Field	Restricted	12	12

A. Multilayer Voting Performance and Ambiguity Filtering

The primary objective of the two-stage voting scheme is to successfully identify correct stars while robustly filtering out false positives caused by sensor noise, hot pixels, or ambiguous geometric star-pairs. During the evaluation, the system extracted the top 20 brightest centroids from the processed images to initiate the primary matching phase.

In Test Image 1 (Clean, Moderate FOV), the pipeline operated under optimal conditions. Out of the 20 primary star candidates tracking through the database, 18 successfully accumulated enough relational matching votes to pass the secondary mutual-consistency verification stage. This high survival rate indicates excellent geometric consistency with the reference Hipparcos catalog.

Test Image 2 acted as a crowded wide-FOV stress case. Due to the large field of view, the background thresholding initially captured a massive profile of 3,283 candidate pixel groups, heavily plagued by structural noise. Despite this extreme clutter, the multi-layer algorithm confidently isolated 8 high-confidence stars that perfectly matched the mutual distance constraints, successfully discarding thousands of false noise points.

Most critically, Test Image 4 validated the safety boundaries of the system. This image represented a sparse star field where only 12 centroids were available. While the primary voting stage attempted to force match these targets to catalog IDs, the secondary verification layer detected that these pairs did not

share valid mutual geometric distances. Consequently, zero stars achieved the required secondary vote threshold ($V_2 \geq 2$). Rather than passing corrupted data forward, the system successfully triggered a "Safe Rejection" loop, preventing the calculation of a compromised attitude matrix.

B. WCS Coordinate Validation

To independently validate the voting accuracy, the accepted pixel centroids were converted to sky coordinates using external WCS astrometric solutions and compared against the assigned Hipparcos vectors.



Fig. 2. Annotated high-confidence matched stars cross-verified successfully against WCS sky coordinates.

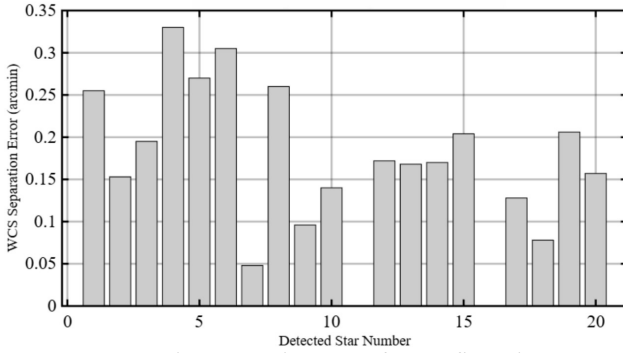


Fig. 3. WCS angular separation error for confirmed stars.

For all solved images, 100% of the high-confidence stars were successfully WCS-confirmed. Mean separation errors were exceptionally low (0.190, 0.334, and 0.519 arcmin respectively), falling well beneath the 5 arcmin maximum tolerance threshold

C. Attitude Estimation Consistency

The confirmed correspondences were passed to both QUEST and Davenport q-method solvers. For the solved test cases, the quaternion outputs generated identical right ascension and declination vectors, resulting in a 0.00000000° angular difference between solvers. This highlights extreme

mathematical consistency and proves that the multilayer verification successfully isolated clean data.

IV. CONCLUSION

This study successfully developed and implemented a Lost-in-Space (LIS) star identification and attitude determination system on a low-cost Raspberry Pi 5 embedded platform. By integrating a multilayer geometric voting framework with a secondary consistency verification stage, the pipeline effectively mitigated false matches and produced highly reliable, WCS-validated star correspondences. Subsequently, QUEST and Davenport algorithms generated consistent quaternion-based orientation solutions. While current limitations include a reliance on pre-recorded images under stable conditions, this framework establishes a practical foundation for CubeSat navigation. Future work will prioritize runtime benchmarking, mitigating dynamic motion blur, and integration with live camera modules.

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